

Ashish Meena

Junior Undergraduate
Department of Computer Science Engineering

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📍 Ashish Meena | 💻 Ashish Meena

ACADEMIC QUALIFICATIONS

Year	Degree/Certificate	Institute	CPI/%
2021 - Present	B.Tech	Indian Institute of Technology Kanpur	7.61/10
2021	CBSE (XII)	MVN Aravali Hills, Faridabad	95.20%
2019	CBSE (X)	D.A.V Public School, Faridabad	96.00%

PROJECTS AND ASSIGNMENTS

- **VolksMarkt** (CS253 Project, Prof. Indranil Saha, CSE, IITK) 📍 (Jan'23 - Apr'23)
 - Built a **full-stack web app** that aims to **digitize** various physical stores and service providers at IITK campus
 - Developed the web-application using **ReactJS**, **Python** with **Django** and **SQLite** in a plan-driven manner
 - Requirements and design documents were generated followed by implementation, **beta testing** and bug fixes
- **CSE Bubble: A MIPS CPU** (CS220 Assignment, Prof. Urbi Chatterjee, CSE, IITK) 📍 (Jan'23 - Apr'23)
 - Developed a **Verilog** description of **CPU** capable of executing a subset of instructions of the **MIPS ISA**
 - Supported **single cycle** execution of **25** different instructions including arithmetic, branch and jump
 - Wrote **MIPS** code for **bubble sort**, converted it to machine code, and simulated its execution on CSE-Bubble
- **Cervical cancer screening** (CS253 Assignment, Prof. Indranil Saha, CSE, IITK) (Apr'23 - Apr'23)
 - Built a **binary classifier** for cervical cancer screening using a dataset with 32 risk factors and 4 target variables
 - Utilized **Python**, **numpy**, **pandas**, **scikit-learn**, and Jupyter notebooks for data analysis and model building
 - Employed feature engineering, data cleaning, and model selection techniques to achieve **91.40%** accuracy
- **Space Travel System** (CS253 Assignment, Prof. Indranil Saha, CSE, IITK) (Mar'23 - Mar'23)
 - Designed a space travel system for booking tickets from a source planet to a destination planet using **OOPS**
 - Added functionality for **booking**, **deleting**, and **updating** tickets, as well as viewing ticket information
 - Built in error and exception handling to ensure robustness in handling unexpected inputs and user errors
- **Nearly Balanced BST** (ESO207 Assignment, Prof. Surendar Baswana, CSE, IITK) (Oct'22 - Oct'22)
 - Implemented nearly balanced Binary Search Tree with logarithmic **insert**, **delete**, and **search** operations
 - Added **self-balancing** mechanism to maintain tree balance, even with skewed element insertion
- **Demystifying Algorithms** (Programming Club, IITK) (May'22 - Aug'22)
 - Gained knowledge of basic data structures such as vector, pair, set, map, queue, stack and priority queue
 - Studied binary exponentiation, **Chinese remainder theorem**, and Fibonacci number in logarithmic time, as well as advanced tree and graph algorithms like BFS, DFS, **Dijkstra's**, **Prim's** and **kruskal's** algorithm
 - Participated in various contests on platforms like Codeforces, Codechef and AtCoder
- **WebDev and UNIX Command Line** (ACA, IITK) (May'22 - Jul'22)
 - Gained experience in version control with **Git** and **GitHub**, and proficiency in using the UNIX terminal
 - Learned frontend development with **HTML**, **CSS**, and **JavaScript** and created a personal homepage

PROGRAMMING ACHIEVEMENTS

- Highest rating **1772(Expert)** on Codeforces(431jericho) and **1909(4 star)** on Codechef(431jericho)
- Highly active on **InterviewBit**(431jericho) with score over **100,000** and ranked under **100** among 60,000 users

TECHNICAL SKILLS

Programming: C, C++, Python

Exposure: HTML, CSS, JavaScript, MIPS Assembly, Bash, Verilog

Software/Libraries: Git/GitHub, Numpy, Pandas, Matplotlib, Anaconda, ReactJS, Autocad, Xilinx, L^AT_EX

RELEVANT COURSES

Data Structures and Algorithms Fundamentals of Computing Introduction to Electronics Linear Algebra and Differential Equations Real Analysis and Calculus	Software Development and Operations Computer Organization Probability for Computer Science Logic for Computer Science Introduction to Economics
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EXTRA-CURRICULAR ACTIVITIES

- Secured **3rd** position in Inter DAV Delhi & NCR chess tournament 2017 in Under-19 category
- Secured **2nd** position in Inter DAV Anand Utsav chess championship 2016 in Under-14 category